

ActInf GuestStream 103.1 ~ Tobias Plowman "The Mind in Motion": A Systemic Model

GuestStream | Tobias Plowman : A Systems Model... | 1:18:54

Hello, welcome everyone.

It is April 21st, 2025.

We're in Active Inference Guest Stream number 103.1 with Tobias Plotman discussing the mind in motion.

So, we will have a presentation followed by a discussion.

If you're watching live, feel free to write any questions in the live chat.

Otherwise Tobias, thank you for joining.

Looking forward to the presentation.

Go for it.

Hey, thank you, Daniel.

So hi everyone, I'm Tobias and this is the mind in motion.

Today I'll be sharing with you a model that I've spent the last few years developing, the neurocognitive systems model or NCSM.

Before we go any further, I just want to do a little disclaimer, just saying I developed this framework independently outside of traditional academic channels, which did allow for novel insights into how cognition dynamically organizes and reconfigures itself.

While this model does explain how cognition works, it also clarifies some systems sometimes fail and sometimes they break down under pressure.

I encourage critical evaluation and I invite you all to consider how this might connect or contrast with your own research.

So what is NCSM?

NCSM proposes cognition as a recursive adaptive control architecture, not fixed or linear.

It's dynamic, precision-weighted and regulated under varying constraints.

The model comprises three integrated layers.

MPPE, the Neurocognitive Predictive Processing Engine, is the generative core predicting and updating cognitive states.

We then have CSTP, the arbitration layer, managing transitions between cognitive states.

And then we have the dynamic cognitive systems or DCSM.

These are five cognitive attractor modes, instinctual, social, executive, memory-driven and associative.

Cognition here is framed as an ongoing inference over internal states regulated by dynamic precision and adaptive control.

At the end of this presentation, you will have a mechanistic account of cognition as a recursive, hierarchical, and dynamically coupled system.

A generalizable architecture grounded in neuroscience, cognitive psychology, and computational logic.

A framework designed for structural adaptability capable of being reconfigured, extended, or embedded.

And a model aligned with the principles of active inference enabling uncertainty minimization through predictive goal-directed dynamics.

So, the way we'll be breaking this down is we'll go through DCSM first, the five cognitive systems.

Then we'll go through CSTP, MPP, and once we've built out the full model, we'll be able to look at some cognitive loops in coherent systems.

Building cognition, behavior, emotion into a mechanical system.

And then we'll go into cognitive breakdowns, dysregulation, and maladaptive feedback loops.

And then we'll get onto some edge case scenarios where we'll look at stuff like addiction, meditation, and some more interesting things.

And finally, we'll end on part seven where we'll look at where this model is going, future work, and collaborations.

So, first we're going to start with the dynamic cognitive systems model.

Sorry.

The dynamic cognitive systems model.

So, there are five key attractive systems, each governing specialized cognitive domains.

First, we have the instinctual core system, which governs reflexes, threat detection, urgency.

Next, we have the social and environmental modulation system.

This balances personal drives against social context.

Next, we have ECHOS, the executive control and optimization system.

This system is responsible for executive control, planning, inhibition, and strategic reasoning.

Fourthly, we have NPRS, the memory pattern retrieval system.

This system is responsible for memory-guided inference using past experiences to inform current cognition.

And finally, we have the associative pattern recognition system.

This system is responsible for symbolic reframing and facilitates creativity, insight, and symbolic adaptation.

Crucially, these aren't isolated modules.

They recursively influence, suppress, and reshape each other in real time.

First, I'm going to introduce the three behavioral systems.

Traditionally, we conceptualize cognition as hierarchical, executive reason on the top, reflexes in the bottom.

Where in reality, cognition is a recursive and multi-directional process.

This interplay lends itself to dynamic precision weighting, where the brain decides moment by moment how much weight to give instinctual cues, social expectations, or executive predictions.

The first system we're going to look at is the instinctual core system, or ICS.

This is cognition's rapid response module and doesn't deliberate, it acts.

ICS is survival first, speed focused, and consistently scanning for threat signals or unexpected changes.

Think reflexes, instinctual urges, immediate emotional reactions like panic or freeze.

Its operational strategy is biased towards action, even if the action might be a false alarm.

From an evolutionary standpoint, it's better safe than sorry.

ICS minimizes immediate bodily prediction errors by quickly aligning organism states to perceived threats, prioritizing urgent action over perfect accuracy.

While ICS is vital, over activation is costly.

Chronic ICS arousal means chronic anxiety, hyper vigilance, or reactive impulsivity, which can cascade into broader cognitive dysregulation.

ICS is essential, but must be carefully regulated.

If relied solely on ICS, there'd be no reflection or any reaction.

That's where the social and environmental modulation comes in.

CEMS take ICS signals and filters them through social context and emotional tone and environmental norms.

CEMS asks, is this behavior socially acceptable or appropriate?

It processes the raw ICS inputs like anger, fear, or excitement, and evaluates their social consequences.

It regulates our expressiveness and ensures coherence with social norms.

CEMS balances authenticity with social coherence, maintaining acceptance and harmony within groups.

In active inference terms, CEMS modulates precision assigned to social signals, emotional expectations, actively inferring what emotional states and behaviors align with predicted social landscape.

CEMS becomes overly restrictive individuals, individuals suppress emotions excessively, resulting in anxiety, isolation, or depressive states.

Conversely, weak CEMS signals lead to inappropriate outbursts or social rejection.

CEMS ensures our instinctual responses remain socially viable.

Without it, coherent social life and group functioning become impossible.

Thirdly and finally of the behavioral systems, we have ECHOES, or the Executive Control and Optimization System.

ECHOES is the cognitive architect.

It strategically plans, actions, inhibits impulses, and evaluates long-term consequences.

It models future states, setting goals and constraints, and balances short-term desires against long-term well-being.

ECHOES is a powerful system, but costly.

It demands extensive cognitive resources, relying heavily on prefrontal processing.

Its computational load makes it vulnerable under stress or cognitive overload.

ECHOES acts as a precision-weighted control policy, minimizing expected free energy by choosing behaviors aligned with coherent long-term predictions.

For instance, deciding between enjoying leisurely time now or preparing for an important future exam.

It evaluates anticipated reward prediction and uncertainty reduction.

When overloaded by stress, fatigue, or inflammation density, ECHOES loses precision.

Decisions become impulsive, erratic, or overly rigid, leading to maladaptive loops or paralysis by analysis.

Healthy ECHOES function differentiates adaptive, purposeful living from chaotic, impulsive existence.

This impairment is central to many cognitive and clinical disorders.

So now we have the three behavioral systems built.

We can look a little bit at how they interact.

Where traditional views often picture cognition as strictly hierarchical, executive control at the top, and instincts at the bottom.

Where in reality, cognition is much more dynamic.

These three behavioral systems, ICS, CEMS, and ECHOES, interact recursively, forming a flexible, bi-directional loop, rather than a rigid hierarchy.

Let's first explore the bottom-up pathway, starting with ICS, the instinctual unconscious system.

ICS rapidly triggers responses like panic or urgency.

It's not subtle, it acts fast to ensure survival, even at the cost of accuracy.

These signals often move up to CEMS, the semi-conscious social and emotional modulation system.

CEMS acts as a filter, evaluating instinctive reactions through a social lens.

It checks what's socially acceptable, appropriate, or shameful.

Consider panic in a social setting.

ICS signals panic.

CEMS evaluates social context.

If CEMS perceives social risk, it might suppress visible panic, redirecting or internalizing the response.

Crucially, ICS and CEMS can directly suppress or even override executive planning in ECHOES.

For instance, an extreme fear or intense shame can disable your ability to plan or make rational decisions in ECHOES.

Now we can look at some of the long-term bottom-up effects.

Where importantly, repeated bottom-up signals shape how ECHOES sees the world over time.

Persistent fear can gradually train ECHOES to expect threats elsewhere, altering long-term cognitive strategies and causing anxiety disorders.

Continuous shame on negative social feedback can restructure how ECHOES evaluate self-worth, resulting in lowered self-esteem or chronic social anxiety.

Now let's look at the top-down pathway.

Where ECHOES, the conscious executive system, shapes CEMS and ICS.

ECHOES models the world strategically, evaluates long-term consequences and inhibits impulsive reactions.

Through repeated inhibition, reframing and habit formation, ECHOES actively retrains CEMS and ICS.

For instance, deliberative exposure to fearful situations guided by ECHOES planning.

Gradually, ICS gradually teaches ICS that certain threats are manageable, eventually reducing instinctual fear responses.

Similarly, conscious reframing, like cognitive therapy, helps CEMS reinterpret social cues while reshaping emotional reactions and responses over time.

Why this system is dynamic and not fixed.

So these interactions aren't static.

Each system, ICS, CEMS and ECHOES, is an attractor pulling cognitive process towards its domain.

Crucially, with constant feedback, they dynamically rewire each other.

ICS is not permanently dominant.

ECHOES isn't permanently in control.

Each can reshape and influence each other, creating cognitive flexibility and adaptability.

So why does this matter?

Understanding this recursive dynamic explains how reflexive or emotional reactions become controllable over time.

It also explains why control breaks down under stress.

In overload, ICS signals become too strong, overriding CEMS and ECHOES.

Most importantly, this explains long-term behavioral change.

Persistent adaptive or maladaptive patterns emerge precisely from this ongoing recursive feedback loop.

By appreciating this dynamic interplay, we can better understand and eventually guide healthy cognitive development, therapeutic interventions and adaptive behavioral change.

So now we have two more systems of the dynamic cognitive systems model.

These next two systems are responsible for memory and creativity.

First of all, we have the memory pattern retrieval system or NPRS.

This is cognition stabilizer and it draws from stored experience and memory traces to contextualize present events and maintain coherence.

The functional importance of NPRS is it anchors cognition in familiar contexts, rapidly retrieving past schemas and patterns to guide real-time interpretation and response.

NPRS reduces uncertainty by supplying stable priors, essentially predictive memories, which help cognitive systems to interpret ambiguous or complex sensory input.

When you recognize a friend in the crowded cafe, NPRS matches visual cues with stored patterns, immediately stabilizing uncertainty and facilitating social engagement.

Yet strong reliance on NPRS can be problematic.

Traumatic memories or past rigid experiences become self-sustaining attractors, creating looping loops such as rumination or avoidance.

Healthy NPRS maintains balance, retrieving memories sufficiently to stabilize cognition without imposing rigid maladaptive interpretations.

NPRS allows cognition to leverage past coherence in uncertain present moments, but it must remain flexible enough to adapt and can recontextualize experience effectively.

Finally, we have APRS, the associative pattern recognition system.

This is cognition's creative reframe engine responsible for insight, creativity and symbolic adaptation.

Functionally, NPRS engages when existing cognitive models fails.

It recombines symbolic associations, generating novel frameworks or explanations to resolve prediction errors and reestablish coherence.

In other words, APRS represents an exploratory inference system, actively generating alternative hypotheses when habitual predictions fail to adequately minimize uncertainty.

When you face an unexpected life crisis, like a sudden career change, APRS activates to explore alternative possibilities or reframing the situation positively.

However, APRS is inherently unstable.

Excessive activation leads to symbolic overload, overwhelming cognition with too many novel hypotheses or symbolic reinterpretations.

This manifests as confusion, existential distress or dissociation.

Healthy cognition balances APRS with echo stability, avoiding symbolic drift while benefiting from novel

reframing.

APRS gives cognition its flexibility and resilience, but requires careful management to avoid cognitive destabilization.

So now we've looked at all five of the cognitive systems.

So the reason I started with DCSM is because this is where the model originated.

So it was intentionally started as a psychological framework.

But then research kind of led me to develop something called the cognitive synchronization transition protocol.

So DCSM described the individual cognitive modes quite clearly, but it didn't fully explain the transitions between them, especially under conditions of uncertainty.

So I was thinking what precisely triggers the instinctual shifts from ICS to CEMS to ECHOs, or how does the brain move from social emotional evaluation into memory retrieval, for instance.

Without understanding these transitions, cognition appeared incomplete, a set of powerful but disconnected states without clear switching rules.

So this critical gap is filled by a new mechanism.

I call it the cognitive synchronization transition protocol or CSTP.

CSTP acts as the arbitration control layer managing cognitive transitions in real time.

Think of CSTP as cognition's real time traffic controller.

It continuously addresses the state of cognitive coherence to determine whether to maintain current cognitive mode or initiate a shift.

It isn't simply turning systems on or off.

It is dynamically monitoring how stable the current cognitive mode is under real time demands.

Signals monitored by CSTP.

Number one is confidence decay.

Is this current cognitive mode losing its predictive reliability?

Next, we have entropy build up.

Is this uncertainty or confusion escalating to unsuitable levels?

And thirdly, we have neural phase desynchronization.

Are these neural signals underpinning our current cognitive state, becoming misaligned or out of sync?

When these signals exceed a certain threshold indicating instability, CSTP initiates a cognitive mode shift to maintain coherence.

CSTP continuously checks whether the system is coherent or starting to break down.

It asks critical questions in real time.

Is our current cognitive state still precise and reliable enough to manage demands at hand?

Are the levels of cognitive tension or entropy manageable or are they becoming critical?

CSTP also has gating transitions.

If predictions remain high and coherence is stable, CSTP actively maintains the current cognitive mode.

But if coherence deteriorates, if precision is sharply declining or if entropy sharply rises, CSTP gates, initiates the transition.

CSTP's active reallocation of cognitive control.

During this gating process, CSTP reallocates cognitive resources and redirects neural synchrony to restore coherence.

This active gating and synchronization ensures cognition remains adaptive, responsible and resilient, rather than becoming overly rigid or chaotically fragmented.

Why is CSTP so crucial?

In short, CSTP ensures our cognition can flexibly adapt to constantly changing demands without losing structural integrity.

It's the dynamic regulator allowing our cognitive systems to reorganize efficiently, preventing maladaptive loops and cognitive breakdown.

Next, that brings us on to the final part of the model, the neurocognitive predictive processing engine, or NPPE.

Now we've seen CSTP manage cognitive mode transitions, but underneath these transitions and the cognitive attractors lies an essential predictive engine.

This is NPPE.

NPPE is the fundamental closed loop architecture that powers cognitive adaptability.

NPPE is the fundamental closed loop architecture that powers the same as the

NPPE.

It continuously generates predictions about incoming sensory signals, evaluates them against reality, and determines how much discrepancies matter.

NPPE is composed of three components.

Number one is the generative model.

This component predicts what will happen next, creating a coherent expectation or hypothesis about the world.

Number two is the error detection system, or EDS.

EDS checks the predictions from the generative model against actual sensory inputs.

Its job is to detect mismatches between expectations and reality.

And then finally, we have the precision weighting mechanism, where PWM evaluates how significant these mismatches are.

It decides if the discrepancy should trigger a change or if it's minor enough to ignore.

[illegible]

So first, the generative model.

The GM creates a prediction, an internal hypothesis about sensory inputs and upcoming events.

For example, the GM might predict a friend's reaction to something you say.

Error detection.

The error detection then compares this prediction with actual incoming sensory data.

What really happens?

If your friend's reaction matches your expectation, everything stays smooth.

If it doesn't, a mismatch occurs.

This is where precision tuning comes in.

The PWM evaluates the mismatch.

How important is this difference?

If PWM decides it's trivial, the system suppresses the mismatch and maintains its existing cognitive model.

If PWM signals it's significant, an update is needed.

CSTP integration.

When PWM signals a significant mismatch, the GM provides its internal prediction model to align better with reality.

If its revision is substantial, large enough to shift cognitive change, it demands CSTP to initiate a cognitive mode shift.

Imagine you expect a friendly greeting, but instead your friend looks upset.

When CSTP identifies this mismatch, PWM flags it as significant.

And CSTP shifts you from casual conversation, CSTP sends into memory retrieval, NPRS or planning echoes to adapt to this unexpected interaction.

While MPPE typically keeps cognition adaptive and coherent.

The system isn't foolproof.

Under intense stress, overload or malfunction, MPPE can still fail in predictable ways.

So there's three major failure modes.

There's high precision weighting.

So belief lock.

PWM is excessively high.

The cognitive system refuses to acknowledge prediction errors.

This creates stubborn beliefs or rigid thinking despite clear contradictions.

Cognition clings to existing models, refusing to update, think of denial or rigid ideologies.

No precision weighting or noise dominance.

PWM is excessively low.

The system sees every small mismatch as significant, making cognition unstable and overwhelmed.

This leads to cognitive chaos, constant anxiety, hypervigilance or confusion.

Nothing feels predictive or trustworthy.

Thirdly, we have GM claps or loss of structure.

The generative model itself breaks down with no clear predictions conform.

In this state, cognition is completely uncertain and disorientated, unable to guess or model reality effectively.

This manifests as severe cognition disruption, dissociation or acute distress.

So why does MPPE matter?

MPPE isn't merely about getting predictions correct.

It's fundamentally about maintaining coherence under stress and uncertainty.

Healthy cognition balances flexibility, updating models with stability, retaining useful predictions.

MPPE manages this delicate balance and its failure represents serious cognitive vulnerabilities.

Improving MPPE failures give us clear insights into clinical conditions, cognitive biases and maladaptive loops, highlighting the value of robust adaptive prediction engine.

So now we're going to look at some cognitive dynamics and feedback loops.

We've seen each layer clearly.

MPPE predicts, CSTP arbitrates and DCSM expressive cognitive modes.

When integrated, these three layers form a powerful recursive cognitive control loop, continuously adapting to cognition moment by moment.

So if you look at the picture here, we've got a rough kind of breakdown of the cycle from input, from external environment or internal states.

There we can see the signal would go through the generative model to be matched against the predictive state.

If there's a mismatch is sent to the precision waiting mechanism, which then checks for error strength, confidence, relevance and resources.

If it's not important, again, it suppresses and exits the loop.

If not, it's flagged as unstable and activates CSTP.

This is where CSTP then checks coherence, phase, confidence, and confidence.

If it's stable, it delays the switch and re-weights the loop.

If unstable, it begins mode arbitration.

This is where the DCSM's modes compete.

CSTP selects the dominant mode and executes the mode and then the cycle repeats.

This is obviously quite a simple breakdown of it and it's a lot more kind of like higher dimensional than that.

But essentially this continuous loop of prediction, arbitration, expression is precisely what makes thought and coherence adaptive.

But more importantly, it also explains where cognition can fail and any disruptions in this recursive cycle can cause instability, maladaptive feedback loops, or coherence collapse.

So we're going to look at emotion now.

Let's discuss how emotion integrates into predictive cognitive system dynamics.

Motion isn't random or disruptive.

It actively biases predictions profoundly, shaping how cognition adapts and selects actions.

Emotions serve three crucial functions within cognition.

Adding weight to predictions, emotions emphasize certain outcomes over others.

Fear prioritizes threat avoidance.

Excitement signals reward.

Motion two is modulating inhibition or activation.

Emotion can either energize or dampen cognitive processes.

Anxiety inhibits actions.

Anger facilitates and anger facilitates assertive behavior.

Number three, emotion guides action viability.

Emotion tags signal which actions feel feasible, socially appropriate, or internally coherent, and which seem impossible or undesirable.

Crucially, emotions are not cognitive noise.

They're precise signal taggers, marking internal states with urgency, social relevance, and subjective significance.

Imagine approaching a stage to speak.

Fear signals the possibility of social rejection, amplifying predictions of negative outcomes, prompting caution or avoidance.

Excitement, conversely, might highlight opportunities for achievement, energizing action towards performance.

By integrating emotional signaling into the cognitive system, MPP, CSTP, and DCSM become finely attuned.

Not just to logical accuracy, but also to urgent, social, and adaptive demands of real-time situations.

Emotion ensures cognition doesn't merely predict accurately.

It ensures predictions matter to our survival, social cohesion, and identity coherence.

Next, we'll look at the insight loop.

This is one of the most fascinating cognitive loops, the insight loops.

This process activates when ECHOES, your executive control system, becomes overloaded.

This system hits a wall or is stuck in repetitive, resource-intensive loops without reaching a solution.

When ECHOES is overwhelmed, cognitive control momentarily relaxes, allowing the associative pattern recognition system to passively activate.

APRS, with its wide-ranging associative power, freely explores symbolic connections without the type of constraints of direct control.

In this state, your memory pattern retrieval system retrieves symbolic traces or subtle hints from past experiences, connecting disparate memories and insights.

Once a promising symbolic pattern emerges, ECHOES can re-engage, adopting the fresh insight generated by APRS and MPRS.

It reorganizes a coherent plan around this newly discovered pathway.

This is how symbolic recombination not only resolves cognitive overload, but revitalizes executive function, providing sudden clarity and those aha moments.

This is how your cognition adapts when a plan fails or when goals become impossible.

This signals growing uncertainty and stress within the cognitive system.

It's actively looking for viable symbolic alternatives.

This is effectively reframing the situation, replanning and correcting your internal narrative.

This is the goal reset.

But what happens when these loops fail or misfire?

Understanding these loops is crucial because they underpin many cognitive and emotional difficulties, like avoidance, denial, or persistent anxiety.

Like here, we're going to look at the avoidance loop, which is when cognition repeatedly predicts failure or threat, leading the system to suppress action or planning altogether.

Second, the precision weighting mechanism tags this predicted failure as highly significant, creating intense anxiety or uncertainty.

Short-term relief validates avoidance, so the next time the task appears, the cognitive system immediately

predicts failure again, restarting the cycle.

Each repetition strengthens avoidance, embedding disengagement as a default cognitive strategy.

Imagine avoiding a challenge at work repeatedly because you anticipate failure.

Each avoidance episode makes the task appear more daunting next time, reinforcing your belief that you can't handle it.

This loop traps cognition, preventing adaptive growth and leading to chronic procrastination, anxiety, and lowered self-confidence.

Next, we'll look at a belief lock loop.

So this is another critical maladaptive loop where outdated or incorrect cognitive models persist despite clear external evidence contradicting them.

First, we have the generative model maintains an outdated or flawed perception about reality.

This could be an innocent self-perception, belief, or expectation.

Secondly, the error detection system correctly flags this mismatch between predictions and reality.

However, the precision weighting mechanism strongly suppresses these error signals, dismissing the mismatch as insignificant.

With no significant mismatch perceived, CSTP never triggers an update or cognitive mode shift.

CEMS conforms to social or emotional pressures supporting the original belief, and ECHO's rationalizes logically, maintained cognitive coherence, reinforcing the outdated belief.

Over time, the cognitive system defends the faulty belief ever more robustly.

Evidence against it is constantly ignored, denied, or rationalized.

Consider someone firmly believing they're incapable of public speaking.

Even after multiple successful talks, they discount the successes as luck or expectations, preserving their original negative self-belief.

Belief lock restricts cognitive flexibility and prevents growth.

It solidifies cognitive biases, fuels demands, I mean, sorry, fuels denial, and maintains rigid patterns to resist change, even in the face of clear evidence.

Finally, we'll look at the rumination loop.

So rumination occurs when cognition becomes stuck, cycling between cognitive models.

Unable to commit firmly to any one attractor state.

Unable to commit to any other factors.

Rather than resolving uncertainty, rumination continually revisits the same thought patterns without productive resolution.

Step-by-step dynamics.

So rumination would start with a triggering event where CSTP detects uncertainty or instability and tries shifting cognitive modes to resolve it.

That results in a failed arbitration where CSTP can't confidently commit to a new cognitive attractor.

No mode feels entirely stable or appropriate.

With CSTP unable to stabilize, APRS, ECHOs, and NPRS continue cycling endlessly.

Each cycle of APRS, ECHOs, and NPRS introduces slight variations, but no cycle is stable or satisfying

enough to hold the loop.

So imagine repeatedly replaying a past conversation.

Each replay is slightly different, evaluating alternatives, feeling unsettled, and never finding a satisfying interpretation to this rumination.

Why does rumination persist?

Because confidence and precision remains unstable.

CSTP never selects a stable cognitive state, creating a persistent cycle without progress.

Rumination isn't productive.

It drains resources, maintains stress, and blocks adaptive cognitive resolution.

Long-term rumination contributes significantly to anxiety disorders, depressive episodes, and chronic stress.

Now we've got the social saturation and aversion loop.

So this is a mode where CEMS fails to align due to perceived overconformity, triggering an identity-preserving inhibition.

Not out of fear of rejection, but of rejection of belonging by default.

So CEMS detects overalignment in the environment.

So essentially this mechanism is where, how can I explain that?

It's like, where, when you realize everyone does something, so then it biases you to do the opposite.

Basically what that means is internally, our self is made by, it's an internal model of ourselves essentially.

And when our internal model is not underweighted, but when it's not established enough, we reject outside social priors.

So identity stabilizes self-hood by selectively accepting new priors, where, when self-priors are strong, adaptation occurs.

When weak, the system resists change to preserve continuity.

So now we can look at applying NCSM to traditional models or to clinical models.

Here we can see that we have common cognitive patterns like freeze loops, over control, social masking, rumination and alienation.

Here you can see the dominant modes that are mapped to and the clinical expression.

These patterns persist not because of static traits, but because the system fails to reconfigure.

Essentially what we're trying to say is NCSM is the foundation from which these clinical patterns emerge from.

And they're not fixed deficits.

So now we're moving more onto the edge case scenarios and kind of more what I think is a bit getting to some of the more kind of interesting stuff.

So here we're looking at addiction, which provides one of the clearest examples of a maladaptive feedback loop where the cognitive prediction loop itself becomes hijacked.

So step by step dynamics of addiction is ICS triggers urgency where ICS triggers stress or emotional discomfort.

Something feels intolerable.

Next APRS rapidly predicts a shortcut to relieve typically a familiar addictive substance or behavior that has

previously provided immediate relief.

NPRS retrieval confirms this prediction based on past relief experiences, reinforcing the shortcut as highly reliable.

The precision weighting mechanism then actively suppresses contradictions or negative consequences, downplaying evidence that the shortcut is harmful or maladaptive.

Crucially, CSTP gets bypassed entirely.

No cognitive reconfiguration occurs.

No evaluation or reconsideration happens.

The result of this sorry is the outcome is predictive coherence.

A consistent internal state is built entirely on false or damaging assumptions.

Cognition feels stable inside, even as it's systematically compromised.

Addiction persists purely because it artificially stabilizes prediction loops, creating coherence that feels right internally, even while causing severe external harm.

Sorry, now we're on edge cases.

My apologies.

Now we move beyond everyday maladaptive loops into extreme cognitive scenarios.

What we call edge cases.

Edge cases are critical tests for our cognitive models because they reveal how cognition behaves under intense and often overwhelming conditions.

Far from anomalies, edge cases are the sharpest stress tests that clarify the limits and resilience of a cognitive system.

ED states represent moments where cognition either transcends typical constraints, such as during intense meditation, or completely collapses as seen in severe addiction or suicide.

These extremes aren't merely outliers, they're valuable for revealing the fundamental mechanics and breaking points of cognitive coherence.

By exploring ED states, we gain insights into how cognitive systems adapt, survive or fail under extreme pressure.

So, suicide.

One of the most severe edge cases is suicide, defined here not metaphorically, but mechanically as terminal cognitive collapse.

The step by steps of terminal collapse.

Initially, prediction confidence drops critically low, so low that the cognitive system can no longer reliably forecast any future future.

CSTP, which normally reroutes and reconfigures cognitive system modes under distress becomes unable to manage or redirect the system.

Symbolic recombination, APRS, and executive planning echoes.

The systems that usually help to reframe or replan disengage entirely, leaving the cognition with no meaningful alternative situation to the scenario.

Sorry.

ICS dominance.

With no coherent alternative available, the instinctual core system becomes fully dominant, driving urgent, desperate action.

Action here isn't primarily emotional despair.

It's the cognitive system attempting an urgent escape from recursive overload and unresolved prediction error.

The cognitive system identifies no path back to coherence.

Suicide emerges as the only viable resolution, an act driven mechanically by precision collapse.

This scenario is not metaphorical.

It is literally a terminal attractor state, a cognitive condition where the system forcibly exists because it cannot resolve the ongoing predictive overload.

By framing suicide this way, we can better grasp its cognitive dynamics, potentially informing a more effective interventions and support systems.

Next, we'll look at meditation where we'll look at the opposite end of cognitive collapse.

In meditation, which is internal suppression of predictive urgency.

During meditation, cognitive load reduces drastically.

The associative memory and patterns via APRS system and NPRS begin symbolic drift, gently recombining experiences without immediate demand.

Precision weighting relaxes, allowing previously suppressed signals to emerge, yet softly enough to avoid triggering reactive states.

Executive control gradually loosens its inhibitory grip, no longer urgently maintaining coherence.

At the same time, instinctual and social systems fade into the cognitive background, bioting emotional and reactive noise.

This deliberate release reduces internal prediction pressure, creating a state where coherence can be safely renegotiated and restructured.

Importantly, meditation is not about achieving complete mental silence.

It's about symbolic cleanup and coherence restructuring.

It's coherence hygiene, not emptiness.

Finally, we'll look at murder.

And back to the opposite end, a more extreme end of the spectrum.

We can look at where in murder where internal coherence is broken.

But instead of withdrawing, the system externalizes the collapse.

This occurs when a powerful rumination loop fuses urgency, ICS, with identity protection, SEMS, misinterpreting the external action as the path to internal coherence.

CSTP fails to manage the conflict.

Unable to reroute or recalibrate effectively, it relinquishes control, allowing ICS to dominate.

Executive inhibition, ECHO's is bypassed or severely suppressed, driven down by the system's desperate attempt to regain stability.

The internal prediction becomes tragically clear.

Unless this external obstacle is removed, internal coherence cannot return.

This act isn't primarily about rage or hatred.

It's about a mistaken cognitive solution.

The system perceives external harm as an unfortunate but coherent pathway back to internal predictability and stability.

Now we reach a critical point, falsifiability.

This model isn't simply theoretical speculation.

It's designed for rigorous testing.

I'm not just describing cognitive phenomena.

I'm trying to set conditions for experimental validation or disconfirmation.

Let's quickly overview three key predictions,

which offer clear testable paths to validate or falsify the system.

So test one, we have the CSTP collapse.

This is the first falsifiable prediction, which focuses on CSTP, the cognitive synchronization and transition protocol.

This model predicts that before observable cognitive breakdown, CSTP will show a measurable loss of neural synchrony between theta and gamma frequencies.

In EEG recordings, we expect to see a signal disruption clearly in the anterior cingulate cortex, a region critical for conflict monitoring and cognitive arbitration.

Behavioral markers.

Behaviorally, this is this desynchronization corresponds to specific symptoms.

Delayed cognitive responses.

Fragmentation of force and decision making and feelings of disassociation or loss of coherent self-awareness.

So how do we test it?

We measure EEG activity specifically for theta gamma coupling while subjects perform demanding cognitive tasks under stress or overload conditions.

We look for moments immediately preceding observable cognitive failures or breakdowns.

Importantly, the prediction can be falsified clearly.

If cognitive breakdown occurs without any measurable theta gamma synchrony loss in the ACC, our model requires significant revision.

Test number two is precision weighting rigidity and rumination.

This predicts that in the high PWM rigidity, which excessively emphasizes existing cognitive predictions will block adaptive shifts between cognitive modes.

When PWM rigidity occurs, executive control echoes excessively dominates, preventing access to flexible symbolic reframing APRS and memory context MPRS.

We propose using fMRI to test for heightened and persistent dorsolateral prefrontal cortex activation, indicating rigid executive dominance.

We combine this with pupillometry, measuring pupil constriction as an indicator of cognitive fixation and reduced exploratory behavior under uncertainty.

Subjects will encounter uncertain or ambiguous scenarios in the scanner requiring, requiring, requiring, requiring cognitive flexibility and reframing.

We look explicitly for neural and psychological and physiological traits that indicate rigid cognitive locking.

To falsify our prediction clearly, we must observe flexible cognitive switching, adaptive reframing and exploratory behavior under conditions of demonstrably high echo activation and rigid PWM conditions.

Finally, we have arbitration fails under social conflict testing.

This third test examines how CSTP handles social dilemmas.

Specifically situations marked by high ambiguity and internal cognitive conflict between executive goals and social emotional modulation.

So, ECHOs and SEMS.

So, we have a clear and specific prediction, which is the model predicts that in high ambiguity social dilemmas, CSTP struggles significantly to resolve conflicts between ECHOs and SEMS.

So, rational planning and social alignment.

This unresolved conflict should produce measurable neural signatures, specifically simultaneous subsystem activation and increased frontal theta activity.

Behaviorally, we expect to see delayed responses, indecision, hesitation and signs of internal cognitive conflict during complex social interactions.

So, for the experimental testing and setup, we will conduct EEG tests while subjects navigate morally or socially ambiguous scenarios designed to trigger ECHOs or SEMS conflicts.

We then analyze neural data explicitly looking for simultaneous subsystem activity, elevated frontal theta oscillations and delayed behavioral responses.

Critically, this model can be falsified if we find consistently rapid cognitive resolution without evidence of subsystem conflict markers or frontal theta elevation, even in highly ambiguous social dilemmas.

Next step.

So, basically, I've been kind of limited in access.

So, I don't really come from an academic background.

So, I've done a lot of internal kind of simulations computationally.

I'm currently working on a runtime environment, which simulates mainly CSTPs core logic of predictive mode transitions.

Arbitration, confidence decay, and stuff like that.

This runtime environment makes cognitive coherence and mode switching not just theoretical, but directly observable and testable.

We can watch cognitive breakdown and adaptive coherence in action, stress test and stress test the mind as rigorously as any control system.

Alongside this runtime, also in very, very early stages, I've developed an internal simulation engine based off NCSM, where I've started to model cognitive processes like recursive breakdown and symbolic recombination.

So, early results are promising and I plan to publicly demo this engine later this year.

Right now, I'm looking to expand beyond solo development.

This model has been, like I said, built entirely independently.

No lab, no degree, no funding.

So, to scale this work, I'm actively seeking collaboration.

If you are running a EEG, MRI or cupilometry lab.

If you are exploring AGI, symbolic architectures or safe arbitration in AI.

If you are involved in clinical diagnostics, psychiatric modeling or neurological conditions.

I would love to collaborate and hear from you.

Just a little bit as well before I finish.

You might be wondering what is SLR and how does it work.

So, SLR continuously tracks not what has changed, but why it has changed.

It then uses that contextual information to decide the next step.

Each decision involves clear logic checks.

Like, did something break?

Is this still coherent and safe?

Should we switch to another cognitive pathway?

Every action within SLR is traceable, transparent and reversible, creating an audit trail of cognitive decision making.

Why does this matter?

Does this matter?

With SLR, we have a symbolic engine capable of simulating complex cognitive phenomena without losing interpretability.

It allows us to study cognition at a granular symbolic level, examining breakdowns, coherence recovery and adaptive stability.

Cognition is more than just function.

It's a continuous recursive act of coherence management.

Your beliefs, your blind spots, your identity itself, all depend on how well your cognitive loops withstand stress.

This model provides the tools to visualize, simulate and potentially stabilize these cognitive loops.

Together we can take this further.

Let's collaborate, test and refine the model in practice.

Thank you.

Thank you.

Thank you, Tobias.

Awesome.

All right.

I'm going to ask some questions from the live chat.

So I'll ask one and anyone else can add, and then I'll ask some other questions.

So Isaac wrote, does this theory help explain how societies function or other systems?

I think so.

[illegible]

really be to have it be a science not just an art or a um ad hoc approach when you say that it's reversible what does that mean does that mean does that mean that specific actions can be reversed in their implication

or that just from a computational perspective you have like logging for all the steps

yeah so um let me talk a bit about personal experience so um i'm diagnosed with um um ptsd from some kind of medical stuff it's not anything um so basically what i realized though is that um my ics like ics responses were involuntary so i'd kind of be cognitively or my echoes would be engaged and um i could see involuntary responses happening and i could over time rewrite them through um echo's engagement through cognitive engagement when these ics responses happen

if you can um i don't know how to explain it but it's active integration like actively focusing your cognitive modes internally can restructure um um other system responses and that and that goes both ways like

if you don't regulate your kind of social um like social systems internally um it can all go off like it's just like um we have

a um a bunch of different states inside and

um um none of them or sorry all of them can take control or can

it's very it's very complex to talk about because though it's also dynamic but um essentially yeah i think they can all kind of reconfigure each other

so there was an unconscious semi-conscious and conscious aspect and is that sort of where the buck stops with cognition with our conscious awareness or how does what we become aware of causally re-enter into these lower level because you describe these systems as attractors so they're kind of like patterns of they're like attractor patterns within the same anatomical apparatus even though they they might hinge more pivotally on on some brain regions or neurotransmitters yeah so how do we become aware of some modes but not other modes when it's the same pieces same molecules same brain regions yeah i understand um

again i'm not too i'm not too sure how um

how echoes is like the conscious or the most adaptable one if that makes sense i've still got a lot of research to do on kind of the subsystems the subsystem dynamics but

um

yeah i mean just just it's an interesting question how with the same pieces in very very slightly differing but importantly differing configurations like phase synchrony give rise to these vastly different cognitive outputs even though there there's a lot of overlap with the parts and the functions um awesome yeah i mean what are your next moves how how do you how can people specifically get more involved yeah sure um so i've started uh a unit called the neurodynamic systems unit and the website is uh should be on the screen ndsunit.org so you can contact me there um but yeah i'm kind of looking for um just anyone kind of interested in either validating the model testing the model or working on um some symbolic AI systems with this uh runtime that will

be available shortly what what will it look like when it's available um it will just be a short a small um package with the core um the way it works is you get nodes that perform like a single action um and then you bundle these nodes together into modules and the modules are kind of like simple folded the way it's coded is actually really drag and drop so um and then you program the the config which is kind of like the lattice state config um which is which nodes activate in which order which modules activate in which order um and then it processes information again through like a symbolic chain where um you know it's quite it's similar to a lot of like um like nested if statements a bit but um is really like I said I'm struggling to kind of talk about it because I don't come from an academic background and um I'm like deep in like some complex stuff so um but it's a lot it's a lot like um some nested if statements I'd say

academia is just like one big nested if statement

yeah I can see that for sure

awesome if you have any last comments feel free otherwise this was great I'm glad you shared it with us and I I look forward to staying in touch and learning more amazing thank you very much

okay thank you Tobias see you bye bye bye

thank you

Thank you.